

AMTI (NMTC) - 2011

RAMANUJAN CONTEST - FINAL - INTER LEVEL

1. Let O and I be respectively the circumcentre and incentre of a triangle ABC . Given $C = 30^\circ$. Let E and D be points respectively on AC and BC such that $AE = AB = BD$. Show that $DE = IO$ and DE and IO are perpendicular to each other.
2. Let N be a $2n$ digit number with digits $d_1, d_2, d_3, \dots, d_{2n}$ from left to right (i.e.) $N = d_1 d_2 \dots d_{2n}$ where $d_i \neq 0, i = 1, 2, 3, \dots, 2n$. Find the number of such N so that the sum $d_1 \times d_2 + d_2 \times d_3 + d_3 \times d_4 + d_4 \times d_5 + \dots + d_{2n-1} \times d_{2n}$ is even.
3. $P_1, P_2, \dots, P_n$ be n points on a circle (in order) dividing the circumference into n equal arcs. Find a permutation $Q_1, Q_2, \dots, Q_n$ of these points such that the sum of the lengths of the path $Q_1 Q_2 + Q_2 Q_3 + \dots + Q_{n-1} Q_n$ is maximum.
4. AA^1 is the median of the triangle ABC . BE, CF are the altitudes of the triangle ABC , cutting at the orthocenter H . The line joining E, F meets BC produced at Q . Show that H is also the orthocenter of the triangle $AA^1 Q$.
5. Let $\frac{p}{q} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{256}$ where $(p, q) = 1$. Prove that p is divisible by 257^2 (257 is a prime).
6. a, b, c are real numbers such that $abc + a + c = b$ and $ac \neq 1$. Find the greatest value of the expression $\left(\frac{2}{a^2+1} - \frac{2}{b^2+1} + \frac{3}{c^2+1} \right)$.
7. $x_1, x_2, \dots, x_n$ ($n \geq 2$) are reals satisfying $\frac{1}{x_1+2011} + \frac{1}{x_2+2011} + \dots + \frac{1}{x_n+2011} = \frac{1}{2011}$. Show that $\frac{\sqrt[n]{x_1 x_2 \dots x_n}}{(n-1)} \geq 2011$.
8. ABC is a scalene triangle. Equilateral triangles ABC_1, BCA_1, CAB_1 are drawn outwards of the triangle ABC .
Prove that
(a) AA_1, BB_1, CC_1 are concurrent (at a point K say)
(b) $AA_1 = KA + KB + KC$.
